

Systematic Review

A Systematic Review of Implementing Multi-Criteria Decision-Making (MCDM) Approaches for the Circular Economy and Cost Assessment

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Abstract: This study advances circular economy initiatives by advocating for the use of Multi-Criteria Decision-Making (MCDM). MCDM methods address the complex multi-faceted aspects of a product or process. They enable conflicting calculations of energy, cost, environmental criteria, and payback periods to be balanced. A systematic critical systematic review and bibliometric analysis were conducted to investigate the contribution of MCDM to the circular economy. The Scopus database was the primary data source reviewed. The geographical distribution, main research sources, and keyword co-occurrences were analyzed across 31 peer-reviewed book chapters, conference papers, and journal articles. The journal Sustainability (Switzerland) had the most publications (4), followed by the Journal of Business Strategy and the Environment and the Journal of Cleaner Production, each with two articles. Recently MCDM has gained popularity as a tool for evaluating the circular economy. This growing interest may be attributed to the complexity of the circular economy, as MCDM effectively balances multiple environmental criteria while integrating evaluations of economic cost and social impact. Criteria are incommensurable as each criterion has a distinct unit of measurement, making it impossible to compare outcomes across different indicators. MCDM is thus an ideal technique for assessing different options by integrating criteria within testable frameworks. However, no established patterns for selecting specific MCDM methods were identified. This is despite some options (e.g., combinations of AHP and TOPSIS) being used more frequently than others. In conclusion, all the studies identified financial factors as the most significant or highly sensitive issue in the transition toward a circular economy.

Keywords: economic evaluation; cost; circular economy; multi-criteria decision-making; MCDM; bibliometric analysis



Academic Editor: Darjan Karabašević

Received: 25 April 2025

Revised: 24 May 2025

Accepted: 27 May 2025

Published: 29 May 2025

Citation: Tighnavard Balasbaneh, A.; Aldrovandi, S.; Sher, W. A Systematic Review of Implementing Multi-Criteria Decision-Making (MCDM) Approaches for the Circular Economy and Cost Assessment.

Sustainability **2025**, *17*, 5007. <https://doi.org/10.3390/su17115007>

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1. Introduction

The circular economy may be defined as a transformative economic system that shifts the traditional linear paradigm of “end-of-life” disposal towards a regenerative model, emphasizing the reduction, reuse, and recycling of materials [1]. Its core objective is to promote sustainable economic growth, enhance environmental quality, and generate social value for both present and future generations. The essence of the circular economy is to shift the current paradigm, encouraging society to develop sustainable business practices [2,3]. The circular economy has been defined as a strategic tool driving sustainable economic

growth while minimizing environmental impact [4]. However, evaluating the circular economy remains a significant challenge. There is currently no consensus on the most appropriate method of assessment. The challenge stems from the numerous criteria and factors required to analyze circular economy practices. These often have different units of measurement, making it difficult to conduct a clear and comprehensive evaluation or to select the most suitable processes or materials with confidence. One of the most widely recognized tools used in analyzing the circular economy in recent years is Multi-Criteria Decision-Making (MCDM) [5–7]. The primary objective of MCDM is to identify the most efficient and effective solution for a given problem by systematically evaluating multiple, often conflicting, criteria [8]. MCDM plays a significant role in advancing the circular economy by providing a structured and systematic approach to prioritize complex alternatives based on multiple, and often conflicting, criteria [9,10].

In psychology, MCDM is a rich field of investigation that explores how individuals choose between options defined by multiple features. These choices are often challenging as people need to decide between different competing options (for example, between option A, which is better than option B in terms of attribute 1, and worse for attribute 2). Research from different areas including cognitive psychology, decision theory, behavioral economics and organizational psychology shows that, when faced with this type of choice, people might take ‘mental shortcuts’—comprehensively described within the ‘heuristics and biases’ framework [11]. For example, when faced with multiple attributes (e.g., cost, quality, convenience), people often simplify decisions by focusing on the most salient or easily comparable criteria [12]. This is called a lexicographic heuristic process. Here, decision-makers might prioritize a single attribute and make their selection based solely on that attribute, disregarding others [13]. This reduces cognitive load (the amount of cognitive resources required to complete the task at hand—in this case, make a choice), but can lead to suboptimal decisions, especially if the prioritized criterion is not the most relevant. Another shortcut is the elimination of aspects, whereby people disregard options that do not meet a minimum threshold for specific criteria until only one remains and is thus chosen [14].

The above-mentioned examples of heuristics can be classed as non-compensatory strategies. These bypass trade-offs intrinsic to MCDM [15,16] because a low value on one attribute is not compensated for by a high value on another attribute. Elimination (non-choice) of an option because of its low value is an example of a non-compensatory strategy, because how this option fares against other attributes does not significantly influence the decision-maker [17]. On the other hand, compensatory strategies lead to the consideration of all criteria values. For example, an option with a low value on one criterion (e.g., cost) can be balanced out by a high score on a different criterion (e.g., quality). An example of a compensatory strategy is the weighted additive rule [13], whereby each option is evaluated, rated and the option with the highest final valuation is chosen. More specifically, the decision-maker assigns a weight to each attribute (e.g., cost, quality) where the weight represents the relative importance of the attribute (e.g., cost being more important than quality). Then, for each option, the decision-maker multiplies the weight of each attribute by the score of each criterion (e.g., the score being high if the cost is low). In essence, the scores capture how well each alternative performs on each criterion. Now these products (weight times score) are aggregated for each option, and the option with the highest product value is chosen.

In the context of MCDM [18,19], compensatory strategies can be considered as more likely to lead to optimal decision-making because they account for all attributes for all options and thus foster decision-making based on all available information. This is embedded within the concept of ‘homo economicus’, or rational decision-makers. Many

results from psychological research suggest, however, that people violate the principles of rational choice [20]. For example, contrary to tenets of traditional theories of rational choice (e.g., Expected Utility and Subjective Expected Utility Theories) [21,22], empirical evidence has shown that (a) the relative preference of any two alternatives is dependent on other alternatives, (b) decision-makers do not have a complete and coherent preference order over all options and (c) decision-makers do not always select the option that is highest in value and are influenced by the decision context [23,24].

Given the above, algorithms have been proposed and tested to aid and foster decision-making. They aim to consider all available information before making a choice. This is crucial in the context of the circular economy, as it is a multi-faceted endeavor involving an evaluation of the quality of multiple attributes across different options. Indeed, decision support systems using multi-criteria approaches like TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) and AHP (Analytic Hierarchy Process) have been shown to influence human decision-making by structuring and clarifying complex choices through systematic ranking [25]. These methods provide structured guidance, allowing decision-makers to weigh criteria (e.g., costs, benefits, or risks) and reduce cognitive load by presenting decisions in a clear hierarchy or a ranked list of options. For instance, AHP aids decision-making through a hierarchical model, breaking down decisions into simple, pairwise comparisons of criteria, facilitating clarity in high-stakes or multifaceted decisions. Moreover, TOPSIS, which is commonly combined with AHP, focuses on ranking alternatives by their relative distance from an 'ideal' solution.

In the study reported in this paper, we emphasize the importance of using Multi-Criteria Decision-Making (MCDM) methods to promote Circular Economy principles, with a specific focus on cost-related factors. By implementing MCDM techniques, organizations and relevant industries can effectively evaluate and prioritize various elements that contribute to circular economy practices. The primary objective of this study is to uncover more sustainable and cost-efficient solutions for implementing circular economy principles.

2. Method

The goal of bibliometrics analysis is to discover the fundamental research structure of a field. It reveals the research trend as a productive body of knowledge [26–28]. This study used bibliometric analysis, which is considered an objective and quantitative method of analyzing and investigating data. This study is structured in two main sections. The first section analyzes previous studies using bibliometric analysis using VOSviewer 1.6.20_exe software. This provides insights into the yearly distribution of publications, source coupling, geographical distribution, and keyword co-occurrence. The data were primarily sourced from the Scopus database [29]. Worldwide, researchers generally agree that there is significant overlap in the content indexed by Scopus and Web of Science, as the search results from both databases are often similar. Scopus offers broader coverage in terms of publication volume and subject diversity [30,31]. The analyses include document co-citation and co-word analysis [32]. In this research, we have reviewed publications from a different angle, namely co-word, co-author, and co-citation analysis, to better understand the use of MCDM techniques in assisting the analysis of the costs of the circular economy.

The second section explores the depth of knowledge on the circular economy, with a specific focus on cost factors integrated with MCDM. The primary objective is to summarize the existing body of work in this area and identify key directions for future research. Furthermore, the study examines the functionality and potential of MCDM as a tool to facilitate the transition to a circular economy.

Systematic Literature Review

There are several different literature review strategies. A systematic literature review collects information in a systematic manner [33]. In this study, we considered meta-disciplinary research by evaluating knowledge across multiple disciplines using quantitative methods, namely co-occurrence networks, citation analysis, and trend mapping. Table 1 shows the keyword search queries we used. Keywords were aligned with the goal of this study. The main keywords were “sustainability”, “Circular Economy”, “cost”, “multi criteria decision making”, “MCDM”, and “multi criteria decision making”. The result of each search identified a variety of articles from each search to select all the target publications. Finally, the search was distributed into eight-part combinations as shown in Table 1. This combination was according to Boolean logic to discover and reveal common publications in the search term. Therefore, all the published conference papers, articles, and chapter books in the circular economy, MCDM, and incorporated into cost were identified.

Table 1. Keyword query search terms used.

NO	Keywords- Query	Number of Articles
1	“sustainability” AND “Circular Economy” AND “cost” AND “multi criteria decision making”	17
2	“sustainability” AND “Circular Economy” AND “cost” AND “multi criteria decision making” OR “MCDM”	20
3	“Circular Economy” OR “CE” AND “COST” AND “multi criteria decision making”	30
4	“Circular Economy” AND “cost” AND “multi criteria decision making”	27
5	“Circular Economy” AND “cost” AND “multi criteria decision making” OR “MCDM” OR “ Multi-criteria decision-making”	31
6	“Circular Economy” AND “LCA” AND “multi criteria decision making”	2
7	“Circular Economy” AND “LCA” OR “life cycle cost” AND “multi criteria decision making”	4
8	“Circular Economy” OR “CE” AND “LCA” OR “life cycle cost” AND “multi criteria decision making”	5

Figure 1 shows the steps for selecting relevant publications. Initially, 136 relevant sources were found. Secondly, the initial search was conducted on literature keywords without considering any time frame limitation. The initial search was conducted in the field of “title-abstract-keywords” and the first publications appeared only in 2019. The extraction of research publications was conducted on 8 January 2025. Publication types were limited to “articles”, “conference papers”, and “book chapters”. Reviews were omitted so that only original relevant studies were included. Furthermore, only publications in English were included.

Next, duplicate publications were searched for and discarded. The remaining 76 sources were reviewed based on their titles, keywords and abstracts to identify research that fully aligned with the scope of the study. This resulted in 43 studies. Finally, a full-text review of all papers was conducted to exclude papers not directly relevant to the scope of this study. This resulted in 31 publications remaining for assessment. This systematic review was conducted in accordance with the Preferred Reporting Items for

Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines. The PRISMA checklist has been completed and is included as a Supplementary File with this submission.

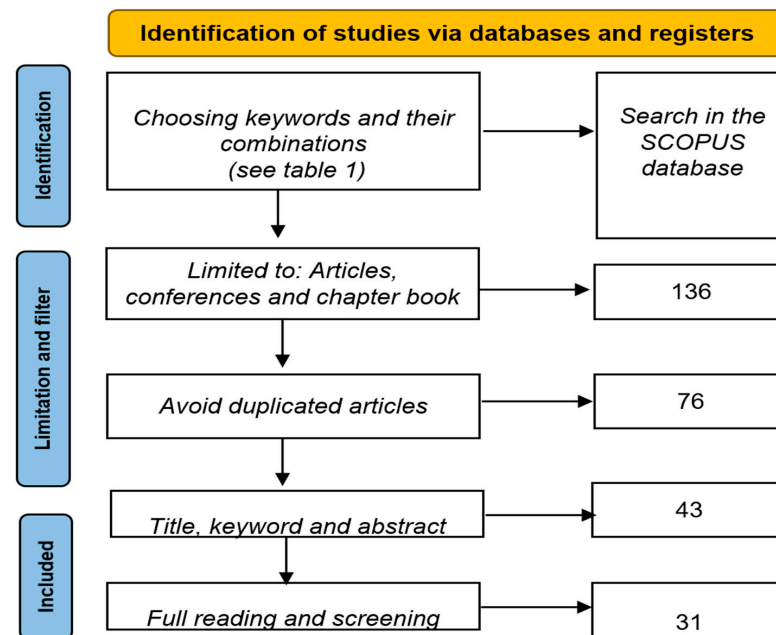


Figure 1. The research process for bibliometric analysis.

3. Results

3.1. Bibliometric Data

No research on the integration of cost, MCDM, and the circular economy was identified prior to 2019. Figure 2 illustrates the trend of publications and citations from 2019 to 1 October 2024. This shows a gradual increase in citations each year, with the number of publications rising from 1 in 2019 to 12 in 2024. This trend indicates a new pattern of integrating circular economy concepts with MCDM in recent years. In this study, we analyze how this emerging topic has rapidly gained attention.

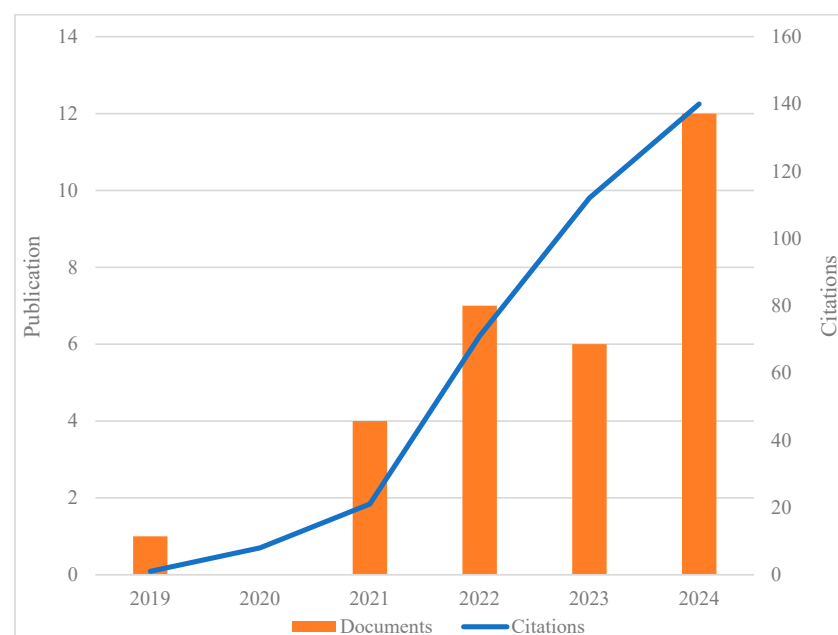


Figure 2. Yearly distribution of publications and citations.

Based on the studies identified, MCDM is primarily used for evaluating various types of waste in the context of the circular economy [34]. The expanding use of the MCDM methodology in waste management can be attributed to the high number of variables involved in finding an optimal option or scenario [35]. It is inefficient to determine the best option based on a single criterion, as this may lead to misinterpretation and suboptimal decision-making strategies [36]. In fact, decision-making without MCDM may be misleading. For instance, Ref. [26] considered three pillars of sustainability—environment, cost, and social factors—in the context of the circular economy. They found that focusing solely on cost may not provide the best solution for industry, as such an approach is fraught with uncertainties. By expanding analysis to include more criteria, MCDM is able to accommodate complexity and mitigate uncertainty, providing more robust and reliable results for decision-makers. The increasing number of publications in recent years corroborates the growing importance of MCDM in circular economy research. This trend underscores the need for comprehensive, multi-faceted approaches to address complex sustainability challenges in the circular economy context.

It can be observed that while the number of publications increased from four in 2021 to seven in 2022, it declined slightly to six in 2023. However, in 2024, the number of publications surged significantly to 12, highlighting the growing importance of this research area. This trend underscores the relevance of the current study and reflects the increasing academic interest in evaluating this field. Furthermore, the study focuses on a research area that has gained attention rapidly in recent years.

Table 2 shows the source of the 31 publications with at least 5 citations. The highest number of papers was published in the Journal of Cleaner Production, Business Strategy and the Environment and Sustainability with 2, 2, and 4 articles, respectively. However, these journals had a relatively low total link strength that shows few contributions to other journals and publishers.

Table 2. Sources of publications.

Source	Documents	Citations	Total Link Strength
Aerospace	1	10	2
Ain shams engineering journal	1	10	7
Biomass and bioenergy	1	11	1
Business strategy and the environment	2	65	18
Discover sustainability	1	5	3
Environment, development and sustainability	1	15	4
Frontiers in environmental science	1	19	8
Heliyon	1	11	6
International journal of production economics	1	46	15
Journal of building engineering	1	4	2
Journal of cleaner production	2	57	6
Journal of urban planning and development	1	9	5
Materials today sustainability	1	9	2
Production planning and control	1	54	21
Sustainability (Switzerland)	4	127	9
Transportation research part e: logistics and transportation review	1	16	4

Technically, a low total link strength indicates weak connections between a journal and other journals based on co-authorship networks and bibliographic coupling. This suggests that the articles published in the aforementioned journals received relatively few citations within the analyzed network.

Another possible reason may be that some journals focus on topics with few connections to mainstream disciplines. In this case, the primary reason for the low total link strength is that most of the journals listed in Table 2 have only a single publication related to MCDM and the circular economy. Meanwhile, journals such as Sustainability, Business Strategy and the Environment, Production Planning and Control, and Cleaner Production received the most citations. This may be because the articles published in these journals are groundbreaking in the field and the area. Studies that merge MCDM with circular economy have attracted a considerable number of citations.

Figure 3 shows the co-citations of sources, with each color representing different clusters of publishers. There are five clusters within the 31 sources of publications. The main purpose of identifying clusters is to highlight publications based on their similarities in terms of co-authorship, citations, and keywords. Thus, dividing publications into clusters makes it possible to identify trends and relationships within a field. Cluster 1 encompasses the Journal of Urban Planning and Development, lecture notes in mechanical engineering, mathematics, Sustainability (Switzerland), and Business Strategy and the Environment. Cluster 2 includes the Ain Shams Engineering Journal, Environment, Development and Sustainability, Journal of Building Engineering, Construction Innovation, and Journal of Construction Engineering and Management. Cluster 3 includes transportation research, logistics and transportation review, frontiers in environmental science, aerospace, discover sustainability, and the Journal of Cleaner Production. Cluster 4 includes the Journal of Environmental Management, Production Planning and Control, the International Journal of Procurement Management, and the International Journal of Production Economics. Finally, cluster 5 includes the Journal of Marine Science and Engineering, Heliyon, Environmental Science and Pollution Research, and Materials Today Sustainability.

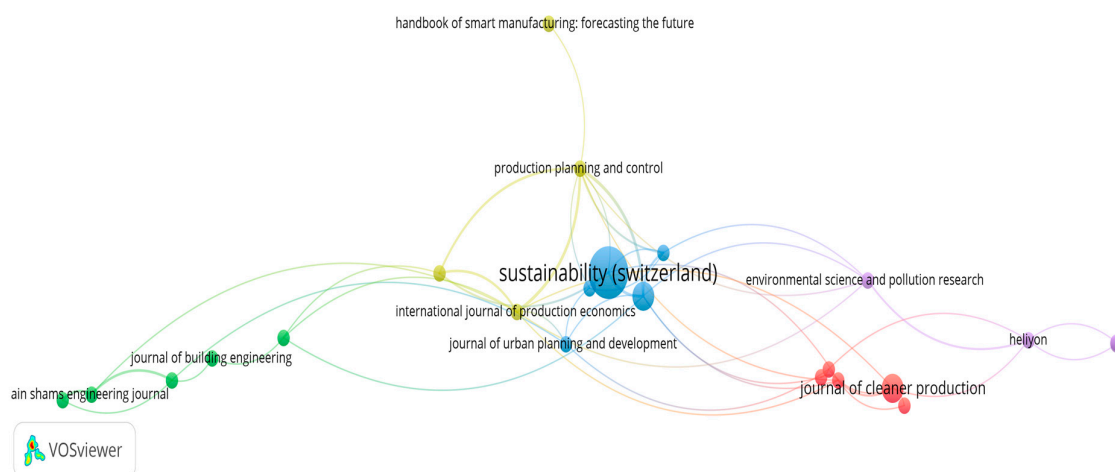


Figure 3. Co-citations of journals and publishers.

In the bibliometric network (Figure 3 generated using VOSviewer), a larger circle represents a higher weight, indicating greater influence within the research field. The size of the circle reflects the impact of a journal, with Sustainability (Switzerland) emerging as a key contributor in the integration of the circular economy, cost, and MCDM. Furthermore, Sustainability (Switzerland) plays a prominent role in this field, as evidenced by its stronger

connections with other journals, reinforcing its influence and significance in advancing research on these topics.

The red cluster's main research focuses are sustainability integration in decision-making, urban systems, and environmental planning. The green cluster's main research focus includes MCDM in civil and building engineering and sustainable construction. The blue cluster is focused on production systems and sustainable logistics. The yellow cluster's focus is on process optimization and environmental evaluation. Finally, the purple cluster's main foci are pollution control and experimental environmental science. Sustainability (Switzerland) emerges as a central and dominant source and key intellectual anchor in the field of circular economy transitions, environmental management, urban planning, and cost modeling.

3.2. Geographical Distribution

Table 3 presents the contributions of various countries to research on MCDM and the circular economy. Several articles include co-authors from multiple countries, reflecting strong international collaboration among researchers and academics working in this field. India, Iran, and the United Kingdom are among the countries with the highest number of publications, with seven, four, and four articles, respectively, and citation counts of 30, 74, and 102. However, the citation distribution highlights a different pattern of contribution. Turkey and Chile, despite having only three publications each, received the highest number of citations, with 110 and 92 citations, respectively. This suggests that the high citation count is associated with the high-impact research output of these countries rather than the sheer volume of publications [37,38].

Table 3. Distribution based on countries.

Country	Documents	Citations	Total Link Strength
Australia	2	25	138
Chile	2	7	61
China	3	92	110
India	7	30	20
Iran	4	74	189
Italy	3	29	291
Malaysia	3	27	197
Serbia	3	37	150
Spain	2	5	4
Turkey	3	110	128
United Kingdom	4	102	199
United states	2	24	109

Surprisingly, the total link strength (TLS) results do not align with the number of citations. Notably, Italy (291), Malaysia (197), and the United Kingdom (199) received the highest TLS scores. This discrepancy may be attributed to the fact that research collaboration (as measured by TLS) does not always correlate with citation impact. While some countries may have extensive research networks and strong collaborations, their publications may not necessarily receive a proportionally high number of citations.

The results indicate that a relatively low number of papers were published in certain countries, such as Chile and Turkey, but received disproportionately high citation counts—110 and 92 citations, respectively. This suggests that their contributions had a significant

impact on the body of knowledge in the field. In contrast, a country like India published seven papers (five journal articles and two book chapters), yet garnered only 30 citations, reflecting a comparatively lower citation performance despite higher publication output.

Figure 4 illustrates the relationship between the number of publications and citations across various countries. It can be observed that Turkey, the United Kingdom, and China received the most citations. Meanwhile, India, the United Kingdom, and Iran have the highest number of publications. This provides insights into both academic impact and research productivity.

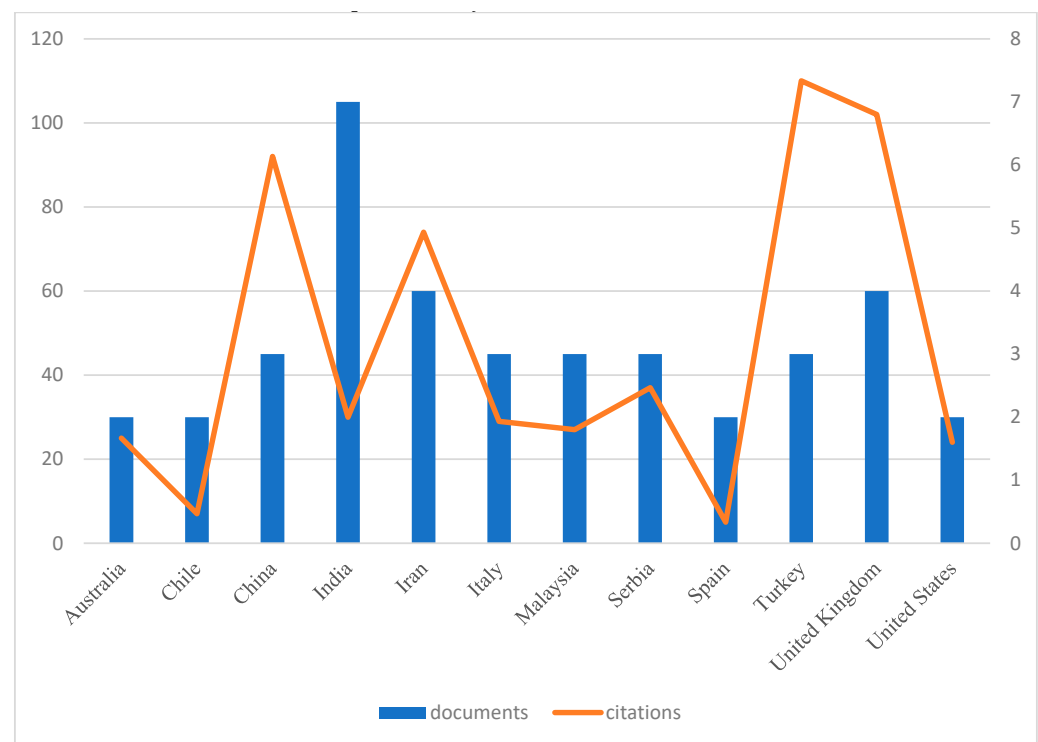


Figure 4. Contribution based on country.

3.3. Keyword Frequency and Co-Occurrence

Table 4 presents the keywords used in the publications as well as their corresponding frequencies. The frequencies indicate the degree of importance of keywords to the rest of the network. Co-occurrence indicates how often each keyword—or certain keywords—appear together across all the selected publications. The preliminary results show that the circular economy has a high frequency, appearing 25 times, and, as illustrated in Figure 5, is represented by the largest circle. However, MCDM (Multi-Criteria Decision-Making) exhibits the highest overall occurrence, totaling 27 instances when all variations of the term are aggregated in the VOSviewer results. This is primarily because many studies refer to MCDM using different formats, such as ‘multi criteria decision making’, ‘multi-criteria decision-making’, ‘multi-criteria decision analysis’, ‘multicriteria analysis’, ‘multicriteria decision analysis’, and ‘multicriteria decision-making’. Despite the variation in terminology, all of these refer to the same concept across the selected publications. A higher co-occurrence of a keyword indicates that the specific term represents a central or core topic within the research field. In the current study, this is evident as MCDM, circular economy, and decision making appear at the top of the list, highlighting their prominence and relevance in the literature. Table 4 shows only those keywords with at list 3 occurrences.

Another important metric presented in Table 4 is the Total Link Strength, which indicates how strongly each keyword is associated with other keywords. For example, MCDM has a high Total Link Strength, meaning it frequently co-occurs with a broad range of other keywords, such as Circular Economy, Decision Making, Sustainability, Analytical Hierarchy Process, and Life Cycle Assessment. This highlights MCDM as a central and highly interconnected topic in the research field, serving as a key point of convergence in scholarly discussions.

Figure 5 shows the keyword co-occurrence network for the selected publications. This shows that the keywords are distributed in five distinct clusters. Each cluster represents a distinct thematic area, highlighting strong conceptual relationships among the associated keywords. Cluster 1, shown in red, primarily focuses on topics such as analytical hierarchy process, cost effectiveness, energy utilization, manufacturing, recycling, and waste management. Cluster 2, displayed in green, revolves around themes such as economic and social effects, life cycle assessment, life cycle cost, sustainability assessment, and sustainable development.

Cluster 3, represented in blue, emphasizes the circular economy, decision making, risk management, supply chain management, and sustainable supply chains. Cluster 4, shown in yellow, highlights comparative analysis and MCDM. Finally, Cluster 5, presented in purple, focuses on the construction industry and modular construction. These diverse clusters reflect the interdisciplinary nature of the field, helping to identify key research directions and understand the interconnections between various topics.

The keyword clusters can be examined further. Cluster 1 is a technical and optimization-focused strand of MCDM applied in sustainable production. Cluster 2 is an evaluative framework of impact analysis and sustainability assessment. Cluster 3 is about systems-level thinking, especially for decision-making and risk under the CE. Cluster 4 focusses on methodologies for comparing or assessing MCDM techniques. Finally, Cluster 5 highlights a sector-specific application area—engineering and building systems.

4. Critical Review

4.1. Circular Economy Analysis

The circular economy involves managing the recycling, recovery, and reuse of materials, as well as reducing the use of raw materials in the process of creating new products. Table 5 shows details of the selected study incorporating MCDM and the circular economy. The primary goal of the circular economy can be described as implementing strategies to reduce environmental impact such as greenhouse gas (GHG) emissions and energy consumption during the manufacturing or construction of new products [39]. However, only assessing the environmental impact of the product does not lead to satisfactory results for stakeholders and consumers [24]. Therefore, cost evaluation must be considered alongside various environmental issues. The challenge lies in the fact that cost is assessed using different units of measurement and is inherently monetary. Thus, researchers need to employ secondary techniques, such as MCDM tools, to conduct a comprehensive assessment of products or processes by appropriately aligning the weighting of different criteria.

One study [40] assessed and ranked alternative mechanical recycling options for plastic waste. They evaluated the integration of two methods based on four key criteria: Economic Factors, Technical Factors, Environmental Impact, and Resource Consumption. It applied [40] rank alternative recycling options based on technical factors, environmental impact, resource consumption, and economic factors within a circular economy framework. They demonstrated that cost is a critical factor in evaluating waste management options within a circular economy.

Reference [41] believed that economic and financial issues were the main obstacles to adopting the circular economy in food supply chains and that MCDM was a great tool to assist in prioritizing the cost challenge. MCDM can assist and ensure that economical constraints are addressed effectively, allowing industry to adopt circular economy trends in a timely manner. Reference [41] used a hybrid MCDM to discover the barriers to the circular economy in the food supply chain. They used Decision-Making Trial and Evaluation Laboratory (DEMATEL) to identify cause–effect, Analytical Network Process (ANP) to prioritize barriers, and finally DEMATEL-based ANP (DANP) to identify interdependencies between barriers. Reference [42] evaluated circular economy alternatives in cacao farming. They evaluated the importance of cost-benefit factors in the agri-food sector to promote the circular economy implementation by considering farmers as key stakeholders. Their study used cost-benefit analysis (CBA), supported by Monte Carlo simulations, showing that drip bean shell tea and irrigation attract the best financial return for stakeholders. Reference [43] investigated using agricultural residues for sustainable building materials. They found that uncertain returns on investment, and high initial investments are among the main obstacles inhibiting transitioning to circular economy practices. Reference [44] evaluated the key strategies for sustainable footwear manufacturing. They evaluated the degree to which expert opinions impact on validating circular economy strategies and on subjectivity influencing decision making.

Reference [45] used a framework to evaluate car-sharing models (i.e., Peer-to-Peer (P2P), Fractional Ownership, Free-Floating, Station-Based, and Carpooling) based on technical (e.g., one-way flexibility), economic (e.g., cost, initial investment and operational cost), social (e.g., accessibility, security and equity), and environmental criteria (e.g., pollution and energy use). Other research [6] developed a Composite Circular Economy Index for EU countries. They found MCDM to be a powerful tool to combine multiple indicators and fill the current gap in existing circular economy measurement. The circular economy has had a strong impact on [36], who argue that neglecting it could suggest entirely different solutions for the building industry.

Some studies encourage MCDM and innovative approaches to business models that enhance circularity. For example, reference [46] used an MCDM tool to evaluate three sustainability pillars, namely economic, environmental, and social factors to enhance decision making accuracy for supplier selection. It claimed [46] that environmental and social factors, along with financial constraints are fundamental criteria for evaluating circular economy frameworks. It proposed [47] a framework for sustainable supplier selection by combining alternative MCDM techniques. They emphasized that cost is the main criterion for selecting a supplier and that the economic efficiency of implementing circular economy practices is one of the main keys for considering financial issues. Consequently, the methodology of combining alternative MCDM tools and cost factors assists organizations to move to circular economy principles. Adopting circular economy alternatives involves a cost-benefit analysis in decision-making. One study evaluated the cost-benefit in the capacity of reused biomass [42].

Table 5. Selected studies incorporating MCDM and the circular economy.

Source	Area of Study	Criteria	Focus of Study
[4]	Risk	Risk of management and decision making, risks related to labor, quality-based risks, design-related risks, performance-related risks, risks related to human resources, supplier-related risks, risks related to material cost, risk of supply chain integration	Supply chain
[5]	Textiles	Environmental, economic, product quality	Sustainability of denim fabric production
[7]	Hybrid Fuzzy Model	Environmental, Technical, Social, Business/Legislative, Economic and Innovation	Contractors in the field of hydropower projects
[29]	Building industry	GWP, human toxicity potential (HTP), acidification potential (AP), terrestrial ecotoxicity (TE), and fossil depletion (FD), and cost	Circularity
[36]	Building industry	Global Warming Potential (GWP), Terrestrial acidification potential (TAP), Human non-carcinogenic toxicity (HCT), Ozone formation, human health (OHH), and fossil resource scarcity (FRS), embodied energy, and cost	Reusability
[37]	Cable and wire industry (recycling key materials like copper and PVC)	Customer, financial, learning and growth, internal process	Circular economy adoption barriers
[38]	Pipe treatments in textile	Low/no-cost high return, (i) high-cost high return, and (ii) medium-cost medium return. Resource efficiency (water, energy, and chemicals), financial feasibility, and environmental impact	Cleaner production and SEMP
[40]	Waste management	Economic, technical, resource utilization, environmental	Plastic waste
[41]	Food chain	Sets of 15 barriers across 6 dimensions (production, management and collaboration, technical and technological capabilities, financial issues, government policies, cultural barriers)	Food waste
[42]	Agricultural	Specialized knowledge required, economic value (e.g., savings or additional sales per kilogram), investment in technology, access to markets. implementation feasibility, biomass efficiency and availability nationwide.	Implementation feasibility, biomass
[43]	Agricultural Residues to a Building Material	Cost (e.g., lack of initial investment), infrastructure (e.g., facilities and specialized equipment), technology (e.g., engineering systems), knowledge (e.g., specialized knowledge, incomplete and imperfect information), policy (e.g., unfavorable and uncertain fiscal policies) an social and cultural (e.g., resistance to change)	Agricultural residues into building materials
[44]	Textile	Nine CE strategies are analyzed (election of low-impact materials and renewable energy, reverse logistics, efficient use of resources, standardization, refurbishment, corrective maintenance, waste material recovery and reprocessing,	Fashion footwear

Table 5. Cont.

Source	Area of Study	Criteria	Focus of Study
[45]	Car-Sharing	Technical (e.g., one-way flexibility), economic (e.g., cost, initial investment and operational cost), social (e.g., accessibility, security and equity), and environmental criteria (e.g., pollution and energy use)	Carpooling
[46]	Supply chain	Economic (e.g., product cost/price, IT facilities), Environmental (e.g., green product design, R&D in environmental issues) Social (e.g., compliance with regulations, work safety procedures)	Agri-based manufacturing
[48]	Fuels for Navy Ships	Density, Autoignition temperature, Flammable limits, etc. Safety, Global Availability, Supply Capacity, Durability, Adaptability, Engine Performance, Engine Emissions, and Cost	Alternative Fuels
[49]	Heavy metals from biomass liquid	Surface area, Porosity, Stability, and Reactivity	Nanocomposites
[50]	Sustainable waste management	Global Warming Potential (GWP), Abiotic Depletion Potential (ADP), Marine Ecotoxicity Potential (METP), Freshwater Ecotoxicity Potential (FETP), Terrestrial Ecotoxicity Potential (TETP), Eutrophication Potential (EP), Terrestrial Acidification Potential (TAP), Photochemical Oxidant Potential (POFP), Ozone Depletion Potential (ODP), and Human Toxicity Potential (HTP)	Polyethylene terephthalate
[51]	Waste management	(1) Economic (e.g., costs of materials, maintenance, and disassembly), (2) environmental (e.g., embodied energy, carbon emissions, recyclability, and reusability and (3) social (e.g., user comfort, safety, and innovation).	Refurbishment of obsolete educational public building stock
[52]	Waste management	Monetary, energy, environmental, material, temporal, efficiency, social dimensions	End of life tires
[53]	Construction and Demolition Waste	Economic, environmental, social	Construction and demolition waste
[54]	Aircraft	technological performance and environmental impact, cost, and circularity	Sustainable Aviation
[55]	Method for locating parking centers	Economic, environmental	Recyclable waste transportation vehicles
[56]	Agricultural	Biomass availability, proximity to existing plants, population density	Rice husk-based electricity generation
[57]	Renewable energy supply chain	Resource management, executive capabilities, recycle, social, cost and environment	Renewable energy chain
[58]	Risk	Supply chain complexity, resource availability and quality, technological challenges	Supply chains
[59]	Building industry	GWP, embodied energy, cost, and social	Construction materials

Table 5. Cont.

Source	Area of Study	Criteria	Focus of Study
[60]	Product-service systems (PSS) design	Engineering characteristics, cost-related (e.g., purchase price, operational costs, investment in new technologies)	Value assessment in product-service systems
[61]	Optimization algorithms	Distance	Ranking of tailings

4.2. Multi-Criteria Decision-Making Methods

Figure 6 presents the fields of study that use MCDM to evaluate alternatives that contribute to the circular economy. Most studies used MCDM to evaluate building materials used in the construction industry [26,36,51]. A previous study [36] used AHP and TOPSIS. Reference [24] implemented AHP and PROMETHEE (preference ranking organization method for enrichment evaluation). Reference [51] used MIVES (Modelo Integrado de Valor para una Evaluación Sostenible) and MCDM to define criteria, indicators, and weights for sustainability assessment, as well as the Delphi Technique to engage experts to refine the tool and reduce biases.

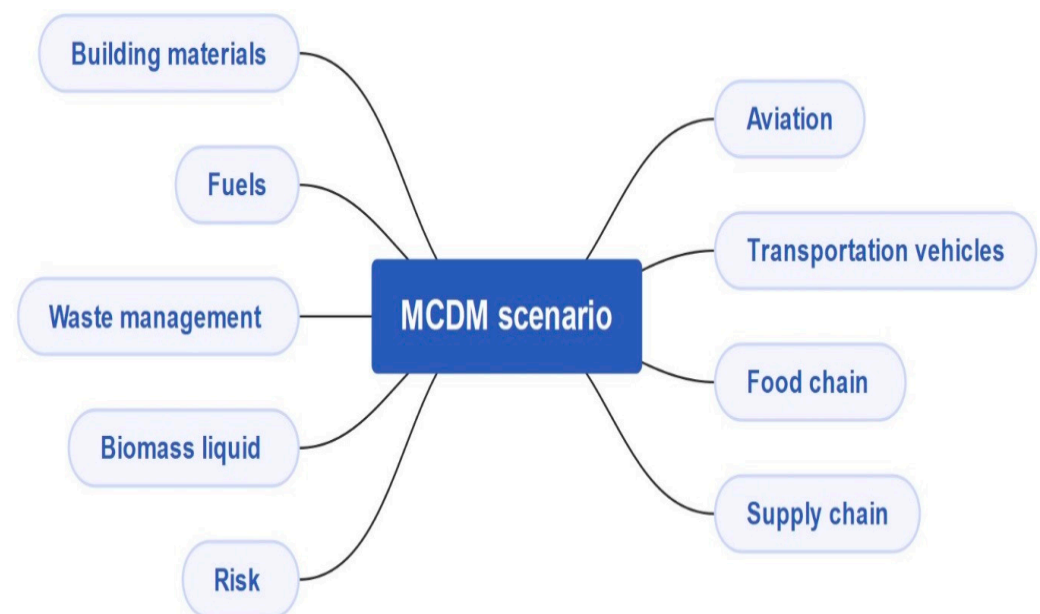


Figure 6. A case study map of MCDM toward circular economy.

Some studies used MCDM on alternative fuels. For example, research [48] used AHP for qualitative analysis. This strongly depends on the accuracy and consistency of pairwise comparisons and is considered subjective and influenced by individual biases.

Other studies have focused on analyzing waste management. For example, study [50] used TOPSIS with AHP (TOPSIS-AHP) and the coefficient of variation (COV; TOPSIS-COV) approaches. A previous study [52] used Fuzzy Analytical Hierarchy Process (FAHP), Fuzzy TOPSIS (FTOPSIS), and Multi-Objective Linear Programming (MOLP). Reference [53] incorporated methods such as bottom-up material stock analysis, cost-benefit evaluation, and scenario analysis. They proposed MCDM to assess different management options (via AHP). Reference [40] conducted studies using the Analytic Hierarchy Process (AHP) and Grey Relational Analysis (GRA), a well-recognized Multi-Criteria Decision-Making (MCDM) approach. AHP was used to assign weights to each criterion based on expert input, while GRA was applied to rank alternative recycling options.

MCDM has also been used in the Aircraft/Aviation industry. For example, one study [54] used AHP to derive the weights of criteria and Weighted Sum Model (WSM) to aggregate normalized scores and thus calculate a Sustainability Index (SI). The results from their WSM study closely aligned with TOPSIS.

MCDM techniques are frequently used in the transportation industry. For example, another study [45] used MCDM to develop a decision framework, combining AHP and TOPSIS. One study [55] relied on an integrated MCDM model that combines DEMATEL (Decision-Making Trial and Evaluation Laboratory) to (1) calculate subjective weights of criteria, then (2) EW (Entropy Weight) to calculate objective weighting and finally (3) WASPAS (Weighted Aggregated Sum Product Assessment) to rank location alternatives. MCDM is also used in the agricultural industry to evaluate the circular economy. For example, reference [42] used simple additive weighting (SAW) as an MCDM tool. Farmers and researchers collaboratively assigned weights on economic value (e.g., savings or additional sales per kilogram), investment in technology and access to markets. Reference [56] used AHP to weigh criteria like biomass availability (A1), proximity to existing plants (A2), and population density (A3). The authors then used the Weighted Linear Combination (WLC) to compute a suitability index by aggregating criteria.

One study [46] combined many MCDM techniques, including Complex Proportional Assessment (COPRAS) to evaluate alternatives, SWARA (Step-Wise Weight Assessment Ratio Analysis) to evaluate expert input, and the entropy method to compute objective weights. Reference [43] applied MCDM by combining the Delphi Method, AHP, and ADAM to prioritize risks in circular supply chains in the agriculture sector.

MCDM has been used in the textile industry to evaluate the sustainability of denim fabric production using recycled cotton fibers and combined heat and power (CHP) plants. One study [5] used an integrated LCA approach combined with the TODIM. Study [44] used FDEMATEL (Fuzzy Decision-Making Trial and Evaluation Laboratory) as an MCDM tool. FDEMATEL is an extension of the DEMATEL technique used in another study [41]. FDEMATEL integrates fuzzy logic to address uncertainty.

A previous study [57] used the CRiteria Importance Through Intercriteria Correlation (CRITIC) technique to assess criteria and the Fuzzy Evaluation Based on a Distance from Average Solution (FEDAS) technique to assess challenges. This study describes an application of the hybrid CRITIC-EDAS approach to assess the challenges of IoT and CE in RESC. One study [58] used a novel hybrid MCDM model that combined the Fuzzy Delphi Method, AHP, as well as the Axial Distance-Based Aggregated Measurement (ADAM) method. The purpose of combining these methods was to ensure uncertainties were addressed.

One research [41] used Decision-Making Trial and Evaluation Laboratory (DEMATEL) to assess the relationships between various barriers and the Analytical Network Process (ANP) to prioritize barriers by assigning weights based on expert evaluations. Reference [43] used the Ordinal Priority Approach (OPA-I) to evaluate barriers and sectors across agriculture, transportation, manufacturing, construction, and operation phases. OPA-I ranks barriers based on expert evaluations using real data matrices. One study [47] used a combination of several techniques such as SWARA, Entropy and COPRAS. Figure 7 shows the frequency of using alternative MCDM in the sources identified for this study. No patterns were apparent for the choice of specific methods, but some methods, such as combining AHP and TOPSIS, were used more often than others.

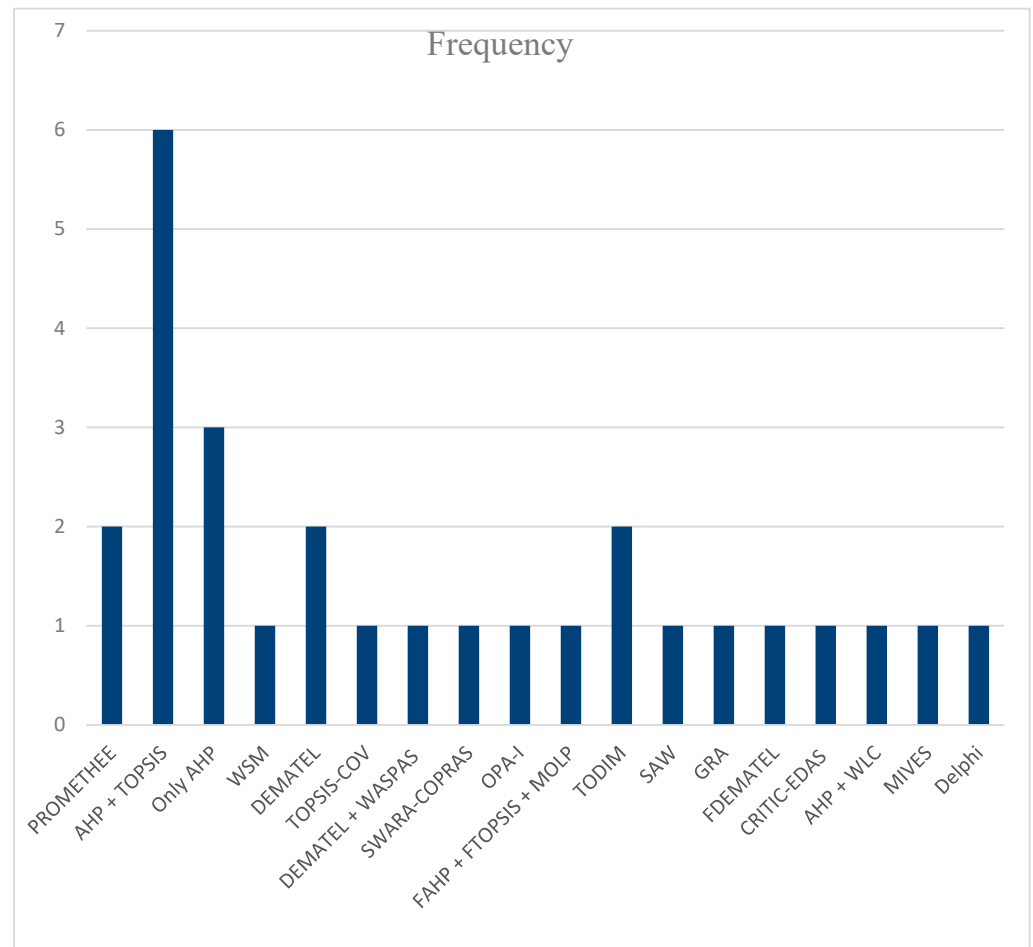


Figure 7. Frequency of MCDM methods.

Different researchers have adopted various alternative MCDM methods in their studies, indicating that there is no consensus on the most accurate or best MCDM method to adopt. Researchers generally select the MCDM method best suited to the problem at hand, the research criteria, the characteristics of the data (qualitative or quantitative), and the decision-makers' preferences. Among these methods, TOPSIS is one of the most widely used in the reviewed studies due to its simplicity in identifying an ideal solution. However, a limitation of TOPSIS is that it does not consider the interrelationships between criteria. It is also possible that using different MCDM methods may lead to discrepancies due to subjectivity in criteria scoring, variations in normalization techniques, and differences in how each method handles weight distribution. Therefore, many researchers choose to combine different MCDM methods in a single study to achieve more robust results. For example, an ideal combination could be to use FDEMATTEL to identify causal relationships [44], subsequently SWARA to determine criteria weights [46], and finally TOPSIS to rank the alternatives [49].

As previously discussed, there is no consensus on the most suitable MCDM method when assessing CE alternatives. However, certain techniques are more appropriate in specific contexts. For example, the AHP is particularly well-suited when expert judgment is required, while the SWARA method is primarily used for determining criteria weights rather than for full decision-making or ranking of alternatives. Future research is needed to investigate the rationale behind the selection of different MCDM methods for various purposes and analyze their comparative advantages and limitations. This should assess alternative MCDM methods and the way they influence results. Furthermore, new research

could also investigate the impact of method selection on decision outcomes and evaluate the robustness and sensitivity of different MCDM methods and whether these change the final results.

4.3. Barriers to Implementation of the Circular Economy

LCA is one of the main and important tools used to evaluate the circular economy. However, it does not consider some goals of products, such as cost savings and quality [62]. Thus, LCA does not provide a comprehensive evaluation of the circular economy. Many recent studies recommend financial models and government incentives as the main solutions that industries can use to overcome the cost barriers and encourage circularity adoption [41,44].

A previous study [52] argues that the barriers toward moving industry to a circular economy include, firstly, consumers' perceptions of recycled products and technical difficulties in processing their end-of-life. Reference [41] believed that the high costs of implementing the circular economy were the second most critical barrier, emphasizing the financial burden of adopting sustainable practices. However, the first critical barrier is lack of circular design and innovative packaging to reduce food waste. Study [44] evaluated the key strategies for sustainable footwear manufacturing. They believed that criteria such as a lack of financial incentives, resource inefficiencies, and high initial investments were among the most important barriers preventing industry from moving towards circularity. In addition, they believed that stakeholder engagement was a fundamental issue in a circular footwear economy.

One study [58] believed that operational costs and high initial investments were among the most important issues and barriers in implementing circular supply chains. One reference [45] found that the high costs of implementing the circular economy were a major barrier. A previous study [63] believed that cost was the main driver and barrier in implementing circular economy principles. One study [37] believed that the highest barriers were high setup costs, lack of financial incentives, poor societal understanding of the benefits of the circular economy (CE), lack of collaboration among stakeholders, and misconceptions about the quality of recycled materials. Reference [57] believed that the most significant challenges of the circular economy included considering the cost of investments, the rate of return on investments, and productivity rates.

Reference [7] created a model for selecting contractors. The model contained six main dimensions: Environmental (EN), Technical (TC), Social (SC), Business/Legislative (BL), Economic (EC), and Innovation (IN). They found that EV and EC were the most prioritized criteria, and included financial viability, resource management systems, and strategies (regenerate, share, optimize, loop, virtualize, exchange). Reference [53] believed that stronger regulations, financial incentives, improved infrastructure, and increased awareness would promote circular economy practices.

Reference [4] identified the risks of transitioning from a linear to a circular economy. There were nine types of risks, namely: management and decision-making risks, risks related to labor, quality-based risks, design-related risks, performance-related risks, risks related to human resources, supplier-related risks, risks related to material costs, and risks of supply chain integration. They found that the barriers during the transition from a linear to a CE for sustainable development included integrated business processes, modular processes for simplification and standardization, and continuous monitoring of cost and performance throughout the supply chain. Reference [40] conducted a pairwise comparison of environmental criteria, resource utilization, economic factors, and technical criteria. They showed that while high-quality recycling initially leads to higher costs, product cost remains

a vital factor for consumers. However, this high cost serves as a major barrier preventing the market from transitioning to recycled products.

The above assessment was conducted based on selected studies in this research that considered cost and MCDM were compulsory to consider in their research. However, other studies, such as the evaluation in [35] that did not consider the MCDM, highlight various barriers to transitioning toward a circular economy. These include a lack of information and awareness, the persistence of economic models based on unsustainable practices, and the absence of adequate incentives for adopting circular practices. In conclusion, all the above studies emphasize the importance of financial factors as the most significant or highly sensitive issue in the transition toward a circular economy.

The discussions above highlight the numerous criteria that can influence the success or failure of implementing the circular economy, including infrastructure quality, consumer perceptions, resource efficiency, operational costs, financial incentives, cost savings, technical challenges, risk factors, regulatory support, and stakeholder engagement. In this context, MCDM methods can assist by ranking the barriers based on their importance, assigning appropriate weights to each factor, and ultimately selecting suitable strategies to overcome these challenges. Given that the circular economy involves evaluating multiple, often conflicting criteria, structured decision-making under complex conditions is required. This is precisely the context in which MCDM methods excel.

Funding from the government or other stakeholders could be used to subsidize the costs associated with implementing a circular economy. Finally, the multidimensional nature of the barriers to adopting the circular economy and the need to simultaneously assess social, economic, environmental, and technical factors clearly demonstrate that the application of MCDM methods is both highly relevant and strongly recommended.

5. Discussion

The process of making informed, balanced decisions in complex situations often relies on Multi-Criteria Decision-Making (MCDM) models. These are designed to aid decision-makers in evaluating alternatives across multiple, often conflicting, criteria. However, while MCDM techniques promise normative rationality, their practical application is fraught with challenges that stem not only from model limitations but also from the cognitive behaviors of decision-makers and the dynamic nature of real-world systems.

One of the core concerns lies in how preferences for different attributes are elicited. Techniques such as pairwise comparisons (e.g., in AHP) or the Best-Worst Method (BWM) are commonly employed. However, the structure of elicitation can fundamentally influence the results. Different methods can produce varying rankings, even with the same underlying data, due to differences in how decision-makers perceive and articulate their preferences. This raises the critical issue of consistency and reliability in preference articulation. Moreover, decision-makers do not always engage with MCDM outputs in a strictly normative fashion (e.g., Tversky & Kahneman, 1974) [11]. Heuristics, cognitive biases, and bounded rationality often lead to selective interpretation or disregard of model outcomes. Despite the development of rigorous normative models, the key question of whether decision-makers use the results in a normative way remains.

Adding to this complexity is the temporal nature of decision-making contexts, which many MCDM models fail to accommodate adequately. Most models offer a static snapshot of preferences and criteria weights at a single point in time. In practice, however, criteria such as costs, environmental impact, and stakeholder priorities can shift due to changing regulations, technologies, or market dynamics. For example, one study [46] proposed real-time optimization techniques as a response to this limitation. Similarly, they critiqued [53] economic models in the context of construction and demolition waste (CDW) management

for failing to account for fluctuating market demand for recycled materials. These examples point to the need for dynamic MCDM approaches that can adapt to evolving conditions.

This temporal challenge also intersects with psychological phenomena such as hyperbolic discounting, where individuals place disproportionate value on immediate outcomes over future benefits. Many sustainability-related decisions, such as those involving circular economy initiatives or lifecycle costing, are particularly vulnerable to this bias. For instance, in evaluating mass timber wall materials, study [9] found that lifecycle costs played a central role in material ranking, highlighting the tension between short-term affordability and long-term sustainability. Other research [55] similarly demonstrated that while optimized waste parking locations reduce long-term transportation costs, they require significant upfront investments—often a deterrent for decision-makers. One study [37] identified initial setup costs as a major barrier to circular economy adoption, reinforcing the need to address temporal trade-offs explicitly in decision-making frameworks. These findings underline a pervasive struggle in aligning short-term decision-making incentives with long-term sustainability goals.

The lack of longitudinal studies exacerbates this issue. While many MCDM applications propose optimal solutions based on current data, few validate whether these choices lead to better outcomes over time. A study [10] call for more longitudinal research to verify the economic and environmental benefits of sustainability-driven decisions. Without such validation, the risk that MCDM models may optimize short-term performance while neglecting long-term impacts remains.

Further complicating the decision-making landscape are methodological challenges related to the structure of MCDM models. One recurring issue is the interdependence of criteria. Some studies [37,55] argue that traditional additive models struggle to reflect complex interactions among criteria. They advocate instead for methods like DEMATEL, which can model causal relationships. Another challenge is the subjectivity involved in assigning weights to criteria. Techniques like AHP and BWM depend heavily on expert input, which may vary across contexts and introduce bias. Reference [50] highlights discrepancies in cost weightings between AHP and more objective methods like the coefficient of variation, illustrating how methodological choices can influence results.

To address subjectivity, some researchers propose integrating subjective and objective weighting approaches. While subjective methods (e.g., SWARA, AHP) rely on human judgment, objective techniques such as entropy weighting offer data-driven balance. Another study [46] demonstrated that hybrid models combining these approaches can yield more robust outcomes—but only if the input data are reliable and of high quality.

Since no single MCDM technique is universally superior to others, it is recommended that future circular economy studies adopt consistent MCDM methods to enable more accurate comparisons and benchmarking across research findings.

In summary, while MCDM models are powerful tools for structured decision-making, their effectiveness is constrained by the methods of preference elicitation, the static nature of their design, and the psychological and behavioral patterns of decision-makers. Future research should focus on incorporating temporal dynamics, validating long-term outcomes, and designing interfaces that align better with real-world decision-making behaviors. Future research may also propose a comprehensive framework or methodology to facilitate cost evaluation for products. This could help and support the industry in accelerating the adoption of circular economy practices.

6. Limitations of Research

This review has offered a comprehensive analysis of the literature on the application of MCDM methods within the context of the circular economy, with particular emphasis on

costs. However, several limitations need to be acknowledged. Firstly, only primary research sources—namely research articles, conference papers, and book chapters—were included, while review papers were deliberately excluded. Review articles typically receive significantly more citations than primary sources. Including them would have disproportionately inflated citation-based metrics such as centrality or importance scores, thereby distorting the evaluation of original contributions. Secondly, we relied on a single bibliographic database, Scopus, to retrieve relevant publications [64]. While Scopus is widely regarded for its high-quality citation data and broad disciplinary coverage [65], the exclusion of other databases, such as Web of Science or Dimensions, may have resulted in the omission of relevant publications indexed elsewhere.

7. Conclusions

The goal of this study is to evaluate the contribution of Multi-Criteria Decision-Making (MCDM) to the circular economy. The significance of this study lies in the fact that circular economy evaluations often involve multiple and distinct criteria such as environmental impact, social implications, and cost assessments. MCDM is a globally recognized tool that provides weighting of each criterion, allowing experts to make informed decisions when faced with numerous alternatives. Therefore, when comparing materials or processes, MCDM tools provide users with a clear and rational choice. Among these criteria, cost evaluation is frequently considered a top priority in selecting the most suitable materials, methods, or strategies. This study builds upon previous research that applied MCDM tools to circular economy practices, with a specific focus on studies that incorporated cost as a key criterion.

The bibliometric analysis offered an overview of MCDM applications within circular economy contexts. The primary results revealed that 31 studies fall within the scope of this research, with a notable increase in publications from 2019 to 2024. India and the United Kingdom contributed the highest number of publications, with seven and four studies, respectively. Keyword occurrence analysis indicated that “MCDM,” “circular economy,” and “decision-making” were the most frequently used terms—highlighting that the central aim of these studies is effective decision-making. The selected papers covered a wide range of industries, including waste management, supply chains, aviation, food systems, building materials, biomass liquids, and risk assessment. This diversity demonstrates the breadth and comprehensiveness addressed in this review. Each study considered a variety of criteria relevant to the circular economy—such as technical aspects of materials, initial investment and operational costs, security and equity, environmental concerns, labor-related risks, and quality-based risks. This further emphasizes the importance of MCDM tools to support well-informed decisions in identifying optimal products or processes. The results also showed that most studies identified cost as the main barrier to implementing circular economy practices.

Future research may propose or recommend a comprehensive framework or methodology to facilitate cost evaluation for products. This could help and support the industry in accelerating the adoption of circular economy practices. One notable issue relates to the variety of the MCDM models investigated. There was little systematic comparison of the different MCDM tools, as most studies adopted one MCDM model or tool. Several studies combined methods like AHP and TOPSIS [50] or hybridized tools such as DEMATEL, WASPAS, and entropy weighting [55] to improve decision robustness. Complex decision problems were handled with novel frameworks (e.g., the fuzzy best-worst method; BWM) integrated with super-matrix structures to prioritize barriers to circular economy adoption [37].

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/su17115007/s1>, PRISMA 2020 Checklist. Reference [66] has been cited in Supplementary Materials file.

Author Contributions: Conceptualization, A.T.B. and S.A.; methodology, A.T.B.; software, A.T.B.; validation, S.A. and W.S.; formal analysis, A.T.B.; investigation, A.T.B.; resources, A.T.B.; data curation, A.T.B.; writing—original draft preparation, A.T.B.; writing—S.A. and W.S.; visualization, W.S.; supervision, A.T.B.; project administration, A.T.B. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Conflicts of Interest: The authors declare no conflict of interest.

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